Customer Churn Drivers and Retention Strategy Analysis in the Telecom Industry

Team 10

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Abstract

Customer churn represents an issue encountered by telecom service providers operating in the environment where customers are highly responsive to prices. Frequent customer changeovers are associated with lost revenues, increased customer acquisition costs, and lost customer lifetime value. Even in such environments where telecom service providers have abundant data from customers, customer churn decision-making happens without understanding the underlying variables that characterize customer churn.

This specific study is intended to learn more about what causes disconnection from telecom service using consumer-level telecom data. The study uses the UCI Telecom

Customer Churn Dataset to establish connections between demographic attributes and billing terms of the clients. The study applies exploratory data analysis in an attempt to find out trends concerning disconnection in order to establish high risk categories among these clients. This study demonstrates how business analytics can be used for decision-making in relation to customer disconnection.

1. Introduction

The telecom industry is characterized by intense competition, minimal service differentiation, and high customer acquisition costs. In such an environment, retaining existing customers is often more cost-effective than acquiring new ones. Customer churn occurs when subscribers discontinue their telecom services due to factors such as high charges, poor service experience, lack of value-added services, or flexible switching options offered by competitors

Even though the telecommunication service providers possess a great amount of information regarding their customers, pinpointing the reasons behind such customers deserting them becomes a difficult task. The analysis of customers' behavior without having any proper information at hand will result in the development of a strategy to retain customers which will be inefficient in handling the underlying causes of customers' discontent. The current project will employ the use of analysis in handling customers' churn to help pinpoint the major causes behind customers' churn.

2. Objectives

The primary objectives of this study are:

- To analyze customer behavior and service usage patterns in the telecom industry
- To identify key factors contributing to customer churn
- To segment customers based on churn risk levels
- To propose data-driven retention strategies to reduce customer churn

3. Team Structure and Roles

3.1 Scrum Master / Project Manager – Sreelakshmi Asok

Responsible for project planning, coordination among team members, timeline management, documentation, and final submission.

3.2 Data Engineer - Valluru Phranav Reddy

Responsible for sourcing the UCI Telecom dataset, data cleaning, preprocessing, handling missing values, and preparing the dataset for analysis.

3.3 Data Analyst – Sanjay S

Responsible for conducting exploratory data analysis, visualization of churn patterns, and identifying key churn drivers.

3.4 Business Analyst - Varun Sivaselvan V S

Responsible for interpreting analytical findings and translating them into actionable business strategies and retention recommendations.

3.5 Machine Learning Engineer - Kallam Jaisimha Reddy

Responsible for applying basic predictive analytics techniques to estimate customer churn risk. This includes feature selection, model building, model evaluation, and supporting customer risk classification, while ensuring model interpretability and business relevance.

4. Data Source

4.1 Primary Data Source

The primary dataset used in this project is the **UCI Telecom Customer Churn Dataset**, obtained from the UCI Machine Learning Repository. This dataset contains customer-level information related to demographics, service subscriptions, billing details, and churn status.

4.2 Key Attributes

- Customer demographics

- Contract type and tenure
- Monthly and total charges
- Services subscribed (internet, add-ons, bundles)
- Churn status (Yes/No)

4.3 Dataset Size

The dataset consists of several thousand customer records with multiple categorical and numerical attributes suitable for churn analysis.

5. Methodology

The case study adopts a systematic process of business analytics. First, there is data cleansing and preparation, which entails missing data handling, categorical variable encoding, and the removal of irrelevant attributes like customer numbers. Finally, there is exploratory data analysis, which seeks to get insights into the nature of customer data.

The crucial churn drivers are determined after examining the correlations between churn and other variables such as contract, tenure, monthly charges, and usage. The customers are then divided into different segments according to their likelihood of churning over. In the final stage, the solutions for customer retention are developed

after deriving useful business insights from the analysis. The approach emphasizes business interpretation over analysing complex models like machine learning.

6. Project plan

Sprint	Week	Activity
Sprint 1	Week 1	Project initiation, understanding telecom business model, and defining the customer churn problem
Sprint 2	Week 2	Identification of business objectives and scope; literature review on customer churn and retention
Sprint 3	Week 3	Dataset sourcing from UCI repository and detailed dataset understanding
Sprint 4	Week 4	Data cleaning, handling missing values, and removing irrelevant attributes
Sprint 5	Week 5	Encoding categorical variables and preparing data for analysis
Sprint 6	Week 6	Exploratory Data Analysis (EDA) to analyze overall churn trends

Sprint 7	Week 7	EDA to study churn behavior across contract type, tenure, and billing patterns
Sprint 8	Week 8	Analysis of service usage and add-on subscriptions influencing churn
Sprint 9	Week 9	Identification of key churn drivers based on analytical findings
Sprint 10	Week 10	Customer segmentation into churn risk categories (high, medium, low)
Sprint 11	Week 11	Design of targeted customer retention strategies and business actions
Sprint 12	Week 12	Business impact estimation, report finalization, and presentation preparation

7. Risk Management

Potential risks include data imbalance between churned and non-churned customers and misinterpretation of correlation as causation. These risks are mitigated through careful data preprocessing, balanced analysis, and clear business-focused interpretation of results.